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Limit Edge Algorithm

Function

A limit-seeking algorithm that dynamically adjusts a limit order based on market conditions such as momentum, volatility, and bid-ask spread to improve fill probability and execution quality while maintaining price control.

- Places limit orders as a market maker to add liquidity at optimal price points
- Monitors market momentum, volatility, and spread to determine order placement
- Adjusts the limit price dynamically to increase execution probability
- Implements a fallback mechanism if the order is not filled within a set time frame
- Ensures controlled execution while maximizing price improvement opportunities

Detailed Explanation

1. Initial Order Placement:

$$P_{initial} = P_{limit}$$

Where P_{limit} is the trader's specified target price

2. Monitoring Market Conditions:

- Momentum (M):

$$M = \frac{P_{current} - P_{previous}}{T_{interval}}$$

- Volatility (V):

$$V = \text{standard deviation of recent prices}$$

- Spread (S):

$$S = P_{bestask} - P_{bestbid}$$

3. Dynamic Order Adjustments:

- Momentum-Based Price:

For buy orders:

$$P_{adjusted} = P_{limit} + (M \times \text{adjustment factor})$$

For sell orders:

$$P_{adjusted} = P_{limit} - (M \times \text{adjustment factor})$$

- Spread-Based Placement:



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- If not filled within time $T_{timeout}$:
 - Make final adjustment within price constraints
 - Revert to original P_{limit} if still unfilled

Real-Life Example

Scenario: A trader wants to buy **5 BTC** with a limit order at **\$26,000**, but the market price is **\$26,100**.

1. Initial State:
- Target Price: \$26,000
 - Market Price: \$26,100
 - Bid-Ask Spread: \$50 (Bid: \$26,075 - Ask: \$26,125)
2. Algorithm Actions:
- Under strong downward momentum:
 - Adjusts order price to: \$26,010
 - When spread tightens:
 - New price = Bid + (Spread/2)
 - New price = \$26,075 + \$25 = \$26,100
3. Outcome:
- Order executes faster than at static \$26,000 price
 - Better execution while maintaining price control

Summary

The Limit Edge Algorithm is ideal for:

- Momentum-Aware Trading:** Dynamically adjusts orders based on price movements
- Optimized Execution:** Places orders inside the spread when favorable
- Failsafe Execution:** Two-step fallback ensures order completion
- Controlled Order Placement:** Maintains price constraints

This algorithm balances execution speed and price improvement, making it effective for enhanced limit order efficiency in dynamic markets.

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